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SUBJECT CODE NO: - RNP-8048
FACULTY OF SCIENCE AND TECHNOLOGY
S.E (EEE / EE / EEP) (NEP) Sem - IV
Examination May/June-2026
AC Machines

[Time: 3:00 Hours] **[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B
1) Use of non-programmable calculator is allowed
2) Q.no.1 is compulsory
3) Solve any four questions from Question No. 2, 3, 4, 5, 6 and 7.

SECTION – A

Q.1 Solve any 10 from the following questions. **20**

1) The V-curves of synchronous motor represent relation between:
A) Speed & torque B) Voltage & current
C) Armature current vs field current D) Power & speed

2) In an alternator, armature winding is placed on:
A) Rotor B) Stator C) Both D) Shaft

3) Cogging in induction motors occurs due to:
A) High voltage B) Low current
C) Magnetic locking D) Overheating

4) In a Synchronous motor, the rotor runs at:
A) Less than synchronous speed
B) Greater than synchronous speed
C) Synchronous speed
D) Zero speed

5) Hunting in synchronous motors refers to:
A) Noise B) Heating
C) Oscillation of rotor D) Overloading

6) The slip of an induction motor at standstill is:
A) 0 B) 0.5 C) 1 D) -1

7) Maximum torque in an induction motor occurs when:
A) Slip is zero
B) Rotor resistance equals reactance
C) Slip is unity
D) Voltage is zero

8) Which starter is used for 3-phase induction motors?
A) DOL starter B) Star-delta starter
C) Auto-transformer starter D) All of these

9) The speed of an induction motor can be controlled by:
A) Rotor resistance B) Stator voltage
C) Frequency control D) All of these

10) The equivalent circuit of an induction motor resembles:
A) Transformer B) DC motor
C) Generator D) Battery

11) The power factor of a synchronous motor can be controlled by:
A) Load B) Speed
C) Excitation D) Voltage

12) Which motor is self-starting?
A) Synchronous motor B) Induction motor
C) Alternator D) Generator

13) The torque in an induction motor is proportional to:
A) Slip B) Rotor current
C) Power factor D) Both A and B

14) Single phase induction motor is not self-starting because of:
A) High resistance B) Low voltage
C) Double field revolving theory D) High current

SECTION – B

Q.2	a)	Explain the torque-slip characteristics of an induction motor.	**05**
	b)	Explain the voltage regulation of an alternator using the EMF method with a neat diagram.	**05**
Q.3	a)	Explain V-curves and inverted V-curves of a synchronous motor.	**05**
	b)	Describe the applications of three-phase synchronous motors.	**05**
Q.4	a)	Discuss the synchronization of alternators and state the conditions required.	**05**
	b)	Derive the torque equation of a three-phase induction motor	**05**
Q.5	a)	Describe the no-load and blocked rotor tests of an induction motor.	**05**
	b)	Discuss the methods of speed control of induction motors.	**05**
Q.6	a)	Explain the construction and working of a single-phase induction motor using Double field revolving theory	**05**
	b)	Describe the different types of single-phase induction motors and their applications.	**05**
Q.7	a)	Explain the alternator on load with resistive, inductive, and capacitive loads using phasor diagrams.	**05**
	b)	Explain the equivalent circuit of a three-phase induction motor.	**05**